

Marketing Campaign Analysis — Phase 2: Data Cleaning & Database Loading

Omar Touzani · Output: marketing_db (PostgreSQL) — 5 tables with PK/FK constraints

I applied all data quality fixes identified in Phase 1, engineered downstream features, and loaded the cleaned data into a relational PostgreSQL database with enforced primary key and foreign key constraints. Campaign rows went from 2,240 to 2,212; reviews went from 28,332 to 25,673.

What changed in this phase

Step	Action	Rows / Cols affected
Constant columns	Dropped Z_CostContact and Z_Revenue	2 columns
Missing income	Dropped 24 rows (see bias note)	–24 rows
Age outliers	Dropped customers with Year_Birth $\leq$ 1900	–3 rows
Income outlier	Dropped customer with Income = \$666,666	–1 row
Education	'2n Cycle' → 'Undergraduate'	Renamed
Marital_Status	YOLO / Absurd / Alone → Single	Consolidated
Reviews duplicates	Removed exact (product, text) duplicates	–1,261 reviews
Short reviews	Removed reviews under 20 characters	–1,398 reviews
Features	Age, TotalSpend, TenureDays, TotalHistCampaignsAccepted, +4 more	+8 columns

Database structure

Table	Rows	Role
customers	2,212	Parent — demographics + tenure (PRIMARY KEY: customer_id)
spending	2,212	Spend per category (FK → customers)
channels	2,212	Web / catalogue / store / deals (FK → customers)
campaigns	2,212	5 historical + Response (FK → customers)
reviews	25,673	Amazon reviews (standalone — no FK to campaign tables)

Important notes for downstream phases

Sample bias from dropped income rows. The 24 customers with missing income had a 4.2% response rate vs 15% for the rest. Dropping them was the right call (small volume, behaviourally distinct), but the remaining dataset tilts slightly toward more engaged customers. The Phase 4 propensity model will be calibrated on this slightly more responsive sample.

Target leakage prevented. *TotalHistCampaignsAccepted* sums only the 5 historical campaigns (*AcceptedCmp1–AcceptedCmp5*). *Response* is the Phase 4 target and must never be included in features that predict it.

Sentiment class balance is still skewed. After de-duplication and short-review filtering, the class split is 89.6% positive / 4.4% neutral / 5.9% negative. Phase 5 will still need class weighting or resampling — evaluation in confusion matrix → classification report → AUC-ROC order, not accuracy alone.

What's next

Phase 3 — Campaign performance analysis. SQL queries against the relational tables to identify which campaigns drove positive ROI, which customer segments responded, and how response patterns differ by demographics and spending behaviour.